Legal Decision Analysis: Plaintiff, Defendant, and Unknown

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Abstract

The research will provide recommendations to legal professionals in litigations related to the appropriate employee versus independent contractor classification issue. The present effort will extend the previous research in this area by employing recent advances in NLP. Specifically, models such as Bidirectional Encoder Representations (BERT) developed by Google in 2018 have shown great potential in solving various NLP tasks with impressive performance. The current project will improve previous investigations in this area — by leveraging the availability of a significantly larger amount of data in the English language from an online repository of US legal documents such as Westlaw and enhanced computing capabilities (i.e., LLMs) to process the readily accessible data.

1 Introduction

Recent NLP applications of LLMs have influenced analytical initiatives in various fields. From large to small, organizations are implementing LLMs to solve various problems (Szilágyi1 and Tóth, 2024). Organizations want to enhance their competitiveness and economic growth through automation, data-driven decision making, and customer service by implementing various LLM applications. Some of the impacts of such implementation have been labeled as transformative throughout the organization through the improvement of efficiency in business processes, innovation, and decision-making. Examples of such applications have been observed in healthcare, finance, education, and software engineering (Pasupuleti et al., 2024). The present research focuses on an application in the intersection of legal and business processes.

According to Mayer (2023), recent integration of AI and LLM based applications promises to revolutionize the research and document analysis in the legal field. Due to these applications, legal professionals are efficiently performing time-consuming tasks such as case law research and contract reviews. Simultaneously, tasks such as document drafting are being performed with greater accuracy and efficiency.

Siino et al. (2025) postulated a framework to encapsulate LLM applications in the legal field. The framework consisted of three broad areas of law, namely, legal search, review of legal documents, and legal prediction. Within each broad area, several legal tasks were included. Legal search included document retrieval, case entailment, and question answering. The second legal area consisted of named entity recognition, similarity estimation, classification, document summarization, dataset extraction, and document automation. Finally, the legal prediction area includes judgement prediction and next sentence prediction.

Our contributions are as follows:

> Preliminary Data Checks and Controls for AnalysisWe finally analyzed 1198 legal documents collected from public repository of Westlaw. Initially, we selected 100 cases to decipher the nature of the data. The initial analysis revealed nine classifications by the judges that included the two main classifications, namely independent contractor and employee. The remaining seven classifications were not purely one of these two categories. The seven were classification 3, 4, 5, 6, 7, 8, 9. Insert classification of the nine categories in a diagram here. For the purposes of the central research question to predict the independent contractor and employee categories, we consolidated the remaining seven categories as the third classification. Therefore, the dependent variable in this study consisted of a trichotomy of employee, independent contractor and a third category. **Created a taxonomy of different types of decisions** as determined by the judges. 9 decisions initially, but eventually it boiled

down to 3 decisions: independent contractor (plaintiff), employee (defendant), and undecided. We built LLMs that can learn from these decisions by analyzing the data in plain texts (i.e., natural language format). We tried to test several hypotheses:

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  - Which LLM (among BERT, ROBERTA, DistillBERT, and LegalBERT) can be fine-tuned to produce the best result from 1000 (approx.) training legal documents?
  - Can an LLM mimic a judge's decision solely from the initial summary of a legal document?
  - Which section (initial summary or description) of a legal document can help an LLM perform the best?
- Can a fine-tuned LLM be deployed on a mobile application to assist legal professionals' initial assessment about the final classification of independent contractor and employee?

2 Related Work

3 Method

The objective of this independent study is to develop a natural language processing (NLP) model capable of predicting court decisions based on the content of legal case documents. By applying NLP techniques, this study contributes to the growing body of research that seeks to identify and explain judgment patterns in classification decisions.

In labor court proceedings, a case is initiated by the **plaintiff**, who asserts their right to be recognized as an employee of a **defendant** company. A central focus of this study is to categorize the types of judgments rendered by the labor court, including:

- rulings in favor of the plaintiff,
- rulings in favor of the defendant, or
- alternative combinations of decisions.

The research involves analyzing raw legal documents, performing data preprocessing, and applying NLP methods to uncover judgment patterns in labor case decisions. Additionally, a mobile application was developed to support legal professionals in their practice.

4 Dataset

A total of 1,198 legal documents were collected from Thomson Reuters WestLaw, a comprehensive online repository of United States legal materials. Following a systematic examination of the pertinent sections within each document, a taxonomy of court decisions was constructed. For the subsequent three-class classification task, the corpus was randomly partitioned into 898 training instances and 300 testing instances.

5 Experiments

5.1 Models

We fine-tuned four large-language models (LLMs). They are as follows:

- **BERT** [**DCLT19**]: A bidirectional transformer model pretrained on large-scale unlabeled text corpora. Its training objectives include predicting masked tokens within a sentence and determining whether one sentence logically follows another.
- **RoBERTa**: An optimized variant of BERT that employs modified pretraining strategies, highlighting that BERT was previously undertrained and underscoring the critical role of training design choices.

- **DistilBERT**: A compressed version of BERT obtained through knowledge distillation, designed to reduce model size, accelerate inference, and lower computational requirements during training.
- **Legal-BERT**: A family of BERT-based models specialized for the legal domain, developed to support legal NLP research, computational law, and legal technology applications. The models were pretrained on 12 GB of diverse English legal text, encompassing legislation, court cases, and contracts, sourced from publicly available repositories.

We implemented our models using the PyTorch framework. Pre-trained models from the Hugging Face library [vWBT+23] were fine-tuned with 898 training samples and 300 test samples. Model performance was evaluated using classification accuracy and the F1-score, with results summarized in Table 1. All fine-tuned models outperformed the random-chance baseline of 33% expected for a three-class classification task.

Model	**Accuracy**	**F1 Score**
BERT	40.00%	
RoBERTa	41.67%	
DistilBERT	40.33%	
Legal-BERT	42.67%	

Table 1: Performance comparison of different transformer-based models.

6 Conclusion

References

[DCLT19] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. Bert: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, pages 4171–4186. Association for Computational Linguistics, 2019.

[vWBT+23] Leandro von Werra, Younes Belkada, Lewis Tunstall, Edward Beeching, Tristan Thrush, Nathan Lambert, Shengyi Huang, Kashif Rasul, and Quentin Gallouédec. Hugging face library: A unified platform for machine learning models and datasets, 2023. Accessed: 2025-08-19.